

On-Average Stability of Multipass Preconditioned SGD and Effective Dimension

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When does preconditioning help or hurt generalisation?

Slides: simonvary.com/colt26.pdf

Setup: Preconditioned SGD

Minimise the **population risk** $f(x) = \mathbb{E}_{z \sim Q}[\ell(x, z)]$ from n i.i.d. samples

$S = \{z_1, \dots, z_n\}$ minimizing $f_S(x) = \frac{1}{n} \sum_i \ell(x, z_i)$ with **multipass** preconditioned SGD:

$$x_{t+1} = x_t - \eta_t \textcolor{teal}{P} \nabla \ell(x_t, z_{i_t}), \quad i_t \sim \text{Unif}\{1, \dots, n\}.$$

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Three sources of **curvature** govern the problem:

$\textcolor{blue}{H}$ Loss curvature - Hessian of the loss $\nabla^2 \ell \approx \textcolor{blue}{H}$, $\nabla^2 \ell(x, z) \preceq \beta H$

$\textcolor{red}{\Sigma}$ Gradient noise - covariance $\text{Cov}_z(\nabla \ell(x, z)) \preceq \textcolor{red}{\Sigma}$

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Question

How does the **excess risk** $\mathbb{E}[f(x_t) - \min f]$ depend on the interplay of **H**, **Σ**, and **P**?

When noise and curvature disagree

In the **well-specified** case, i.e.,

$$\ell(x^*, z) := -\log p_Q(z|x^*),$$

$$\Sigma = H = \text{Fisher information.}$$

Then *natural grad.* $P \approx H^{-1}$ is optimal [Amari '98].

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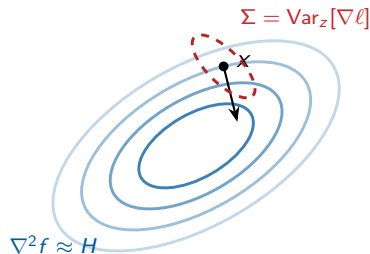
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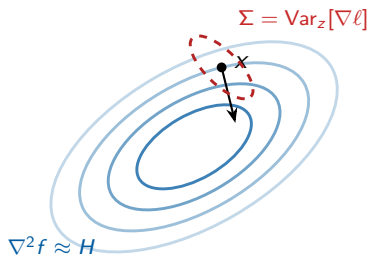
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Whitening the second-moments can destabilise high-curvature directions.



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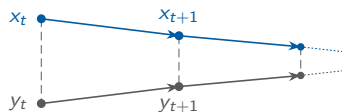
Main result: excess risk of PSGD

Theorem

For contractive updates, step size $\eta_t \sim 1/t^\gamma$, where $\gamma \in (0, 1]$ and $n \gtrsim \kappa_\ell \kappa(PH)$:

$$\mathbb{E}[f(x_t) - \min f] \lesssim \underbrace{\beta \frac{\text{tr}(PHPS)}{t^\gamma}}_{\text{optimisation} \sim 1/t^\gamma} + \underbrace{\text{tr}(PS) \left(\frac{1}{\sqrt{n} t^\gamma} + \frac{1}{n} \right)}_{\text{generalisation} \sim 1/n},$$

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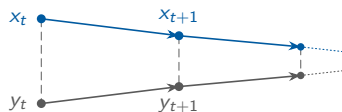
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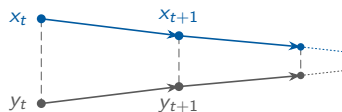
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Optimal preconditioner

$P = H^{-1}$, $\gamma = 1 \Rightarrow$ both terms collapse to

$$\text{tr}(H^{-1}\Sigma) \left(\frac{1}{t} + \frac{1}{n} \right).$$

Recovers the optimal TIC rate.

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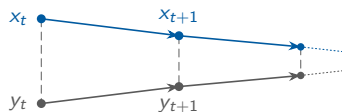
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Second-order information is not only optimization — it is robustness to sampling noise.

Beyond contractivity: At convergence

Theorem

For f_S satisfying μ -quadratic growth (non-convex), and $n \gtrsim \beta/\mu$:

$$\mathbb{E}[f(x_t) - \min f] \leq \frac{2\beta}{\mu} \mathbb{E}[f_S(x_t) - \min f_S] + \frac{16}{\mu} \frac{\text{tr}(H^{-1}\Sigma)}{n}.$$

At convergence the bound is the *optimal effective dimension* — independent of P .

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$P = H^{-1}$ is the *sweet spot*: fastest optimisation and smallest effective dimension.

The right complexity measure: effective dimension

Classical statistics replaces the ambient dimension d by the **effective dimension**

$$\boxed{\text{tr}(H^{-1}\Sigma)}$$

also known as the **Takeuchi Information Criterion** (TIC).

It is the *optimal* statistical rate $\text{tr}(H^{-1}\Sigma)/n$ for:

- exact ridge-regression minimisers [Bach '24],
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This work

We bring $\text{tr}(H^{-1}\Sigma)$ into the *non-asymptotic* analysis of **multipass PSGD** and show how the choice of P can *break* it.

On-average algorithmic stability

Standard approach is worst-case: Say x'_t is trained on $S' = S \cup \{z'\} \setminus \{z_i\}$

$$\mathbb{E}_S[f(x_t) - f_S(x_t)] = \underbrace{\mathbb{E}_{S,z'}[\ell(x'_t, z_i) - \ell(x_t, z_i)]}_{\text{on-average stability}} \leq L \mathbb{E}_{S,z'}\|x_t - x'_t\|,$$

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Instead, to reveal the geometry:

- use β -smoothness (*not* Lipschitzness)
- **on-average** (not uniform) stability $\Rightarrow$ the bound sees the noise geometry Σ .

$$\mathbb{E}_S [f(x_t) - f_S(x_t)] \lesssim \underbrace{\mathbb{E}[\|\nabla \ell(x_t, z')\|_*^2]^{1/2}}_{\text{gradient/noise geometry } \Sigma} \cdot \underbrace{\mathbb{E}[\|x_t - x'_t\|^2]^{1/2}}_{\text{parameter stability squared}}.$$

Stability for *multipass* SGD

Aim: Bound $\mathbb{E}\|x_t - x'_t\|_M^2$ where $\|x\|_M := \sqrt{x^\top M x}$ for $M \succ 0$.

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Lemma

Provided $n \gtrsim \kappa_\ell \kappa(\textcolor{teal}{P}\textcolor{teal}{H})$, the correlated term is benign and unrolls to

$$\mathbb{E}_{S,z'} [\|x_t - x'_t\|_M^2] \lesssim \frac{\text{tr}(\textcolor{teal}{P}M\textcolor{teal}{P}\textcolor{red}{\Sigma})}{n^2} + \eta \frac{\text{tr}(\textcolor{teal}{P}M\textcolor{teal}{P}\textcolor{red}{\Sigma})}{n}.$$

Proof idea: the perturbed iterate has never seen z_i

The two runs differ only when index i is drawn (prob. $1/n$):

$$\mathbb{E}\|x_{t+1} - x'_{t+1}\|_M^2 \lesssim (1 - \eta r) \mathbb{E}\|x_t - x'_t\|_M^2 + \frac{\eta^2}{n} \mathbb{E}\|\nabla \ell(x_t, z_i) - \nabla f(x_t)\|_{PMP}^2.$$

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$$\nabla \ell(x_t, z_i) - \nabla f(x_t) = \underbrace{\nabla \ell(x'_t, z_i) - \nabla f(x'_t)}_{\text{fresh variance} \leq \text{tr}(PMP\Sigma)} + \underbrace{[\nabla \ell(x_t, z_i) - \nabla \ell(x'_t, z_i)] - [\nabla f(x_t) - \nabla f(x'_t)]}_{\text{smoothness: } \lesssim \beta \|x_t - x'_t\|}$$

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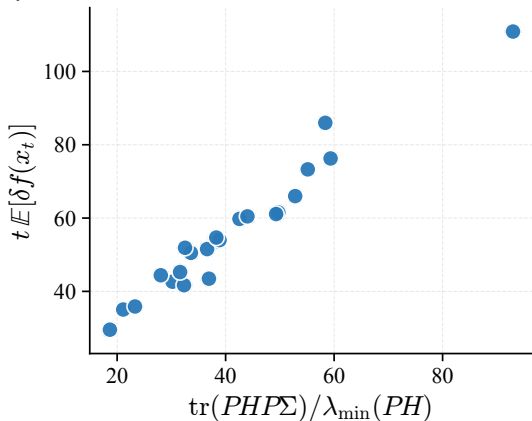
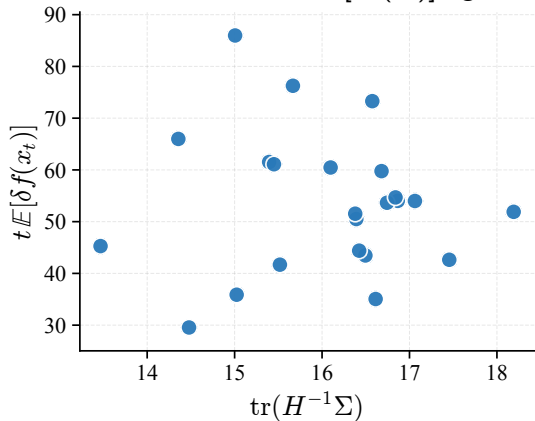
Contraction balances the correlation cost:

$$\mathbb{E}\|x_{t+1} - x'_{t+1}\|_M^2 \lesssim \left(1 - \eta r + \frac{\eta\beta}{n}\right) \mathbb{E}\|x_t - x'_t\|_M^2 + \frac{\eta^2}{n} \text{tr}(PMP\Sigma),$$

So for $n \gtrsim \beta/r$ the contraction absorbs it $\Rightarrow$ same as if all samples were fresh.

Experiment: it's the interplay, not H, Σ alone

Noisy quadratics, 24 random instances (H, Σ, P) (non-commuting), $d = 10$. Plot scaled last-iterate risk $t \mathbb{E}[\delta f(x_t)]$ against two predictors.



Non-asymptotic behaviour is governed by (P, H, Σ) jointly — not by H, Σ alone.

Takeaways

- ① **New tool:** on-average stability analysis for **multipass** SGD handles correlated iterates, no Lipschitz assumption.
- ② **Geometry of generalisation:** excess risk of PSGD is controlled by an **effective dimension** jointly shaped by curvature H , noise Σ , and preconditioner P .
- ③ $P = H^{-1}$ **is optimal** for *both* optimisation *and* generalisation — recovering the optimal rate $\text{tr}(H^{-1}\Sigma)(1/t + 1/n)$, *even under misspecification*.
- ④ **A bad P hurts:** matching **instance-dependent lower bounds** show the constant can blow up by $\kappa(PH)$, even with decaying steps.

Thank you! `simon.vary@stats.ox.ac.uk`

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Matching lower bounds: a bad P really does hurt

Minimax (any estimator). The statistical floor is exactly the effective dimension:

$$\inf_{\hat{x}} \sup_P \mathbb{E}[\delta f(\hat{x})] \gtrsim \frac{\text{tr}(H^{-1}\Sigma)}{n\alpha}.$$

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$$\mathbb{E}[\delta f(x_t)] \gtrsim \frac{\text{tr}(PH\P\Sigma)}{\alpha \lambda_{\max}(PH) \lambda_{\min}(PH)} \cdot \frac{1}{t} + \frac{\text{tr}(H^{-1}\Sigma)}{\alpha} \cdot \frac{1}{n}.$$

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The price of misalignment

- $P = H^{-1}$: matches the upper bound (up to κ_ℓ).
- Ill-conditioned P ($\kappa(P) = 1/\varepsilon$): risk $\gtrsim \text{tr}(H\Sigma)/(\varepsilon t)$ — constant blows up even with decaying steps.
- Plain SGD ($P = I$): a factor $\kappa(H)$ worse than optimal.

Minimax is too coarse here — the *instance* (the choice of P) decides performance.